

Verification of Wein's displacement law and Stefan's law using incandescent bulb

Aim:- To analyse the black body spectrum of incandescent bulb in Tracker Video and to verify the Wein's displacement and Stefan's laws.

Apparatus: Computer with Tracker tool installed and recorded video of the spectrum.

Principle:

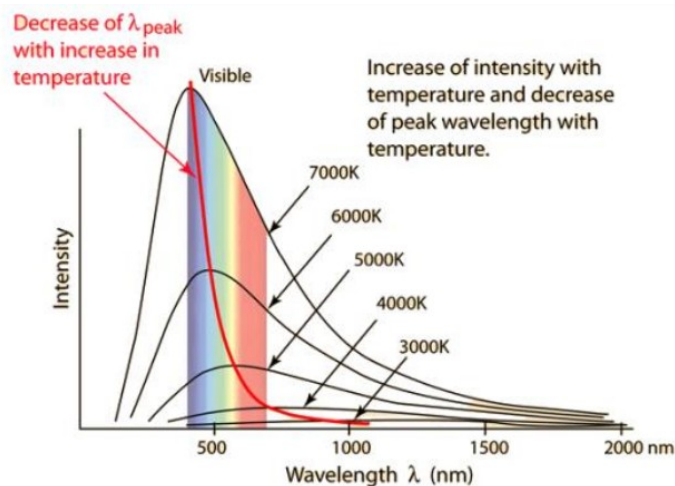
The interpretation of thermal radiation from a black body was proposed by the **Max Planck** in 1900. Based on Planck's result, the intensity of the radiation is given by:

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

This shows that, the wavelength $\lambda_{\max}$ at which the emitted intensity reaches its maximum value decreases as the temperature is increased i.e., $\lambda_{\max} \propto 1/T$ or

$$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m}\cdot\text{K} \quad \text{a constant}$$

This result is known as **Wien's displacement law**. the term "displacement" refers to the way the peak is moved or displaced as the temperature is varied.



The total intensity radiated over all wavelengths (that is, the area under each curve) increases as the temperature is increased.

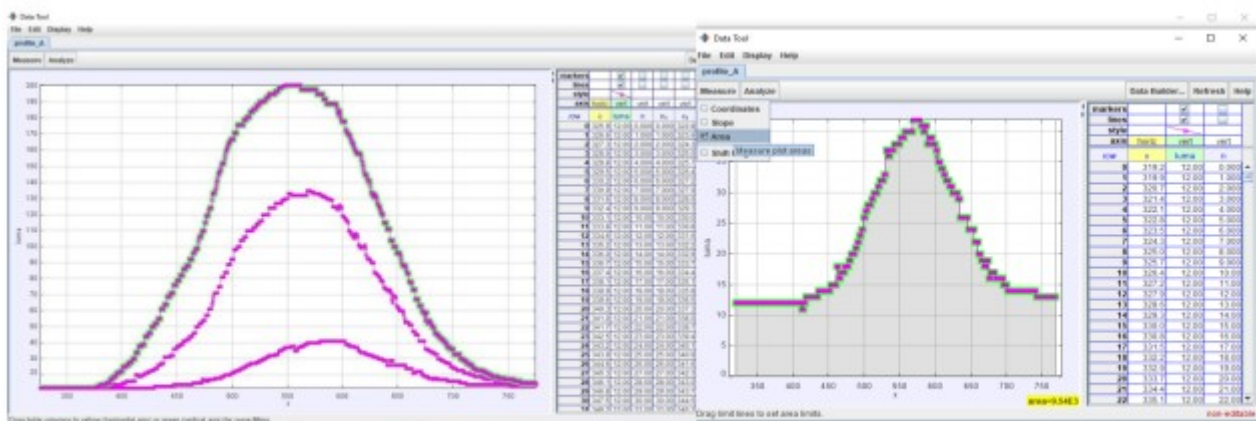
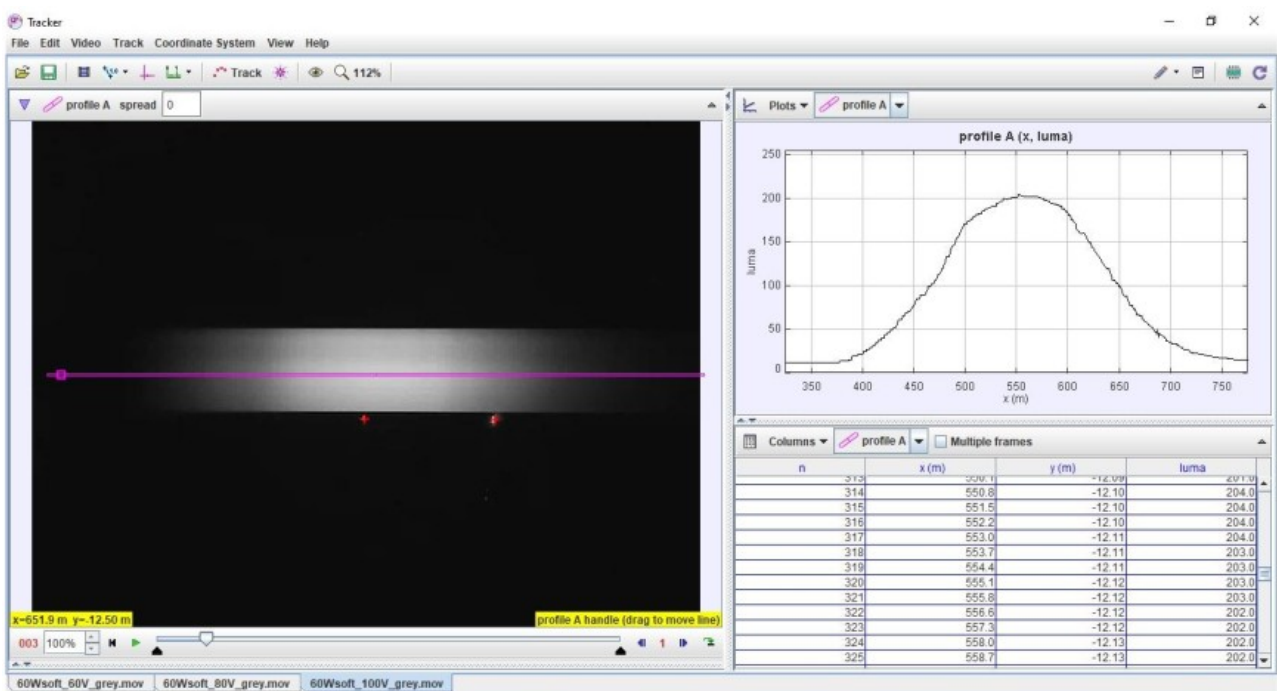
Stefan's law: The total intensity increases as the fourth power of the absolute or kelvin temperature:

$$I = \sigma T^4$$

Stefan-Boltzmann constant $\sigma = 5.67037 \times 10^{-8} \text{ W m}^{-2} \cdot \text{K}^{-4}$

Procedure:

- Download and install tracker video in the computer from <https://physlets.org/tracker/>
- Import the recorded video of the spectrum of incandescent bulb operated at different voltages(hence different temperature) can be used. (For example 60W bulb at 60V, 80V and 100V). (File > Open> File Chooser> select the file).
- Calibrate the spectrum and get the intensity profile as in previous experiments.
- Get the wavelength corresponding to intensity peaks $\lambda_{\max}$ of the spectrum by analysing the tabulated values on the bottom-right column.
- To estimate the area under the curve, right-click on the graph , select analyse Measure> Area
- Draw the profiles one by one and note the $\lambda_{\max}$ and the area under the curve in each case.
- Wein's displacement law can be verified from $\lambda_{\max}$
- Stefan's law can be verified from the area under the curve.



Wein's displacement law

Stefan's law- area under the curve

Observations:

Voltage(V)	$\lambda_{\text{max}}(\text{nm})$	Area Under the curve
50		
60	575.5	9500
70		
80	566.6	24000
90		
100	554.8	39200

Results:

1. As Voltage increases the temperature of the filament of the incandescent lamp increases but λ_{max} decreases. This verifies Wein's displacement law
2. As Voltage increases the temperature of the filament of the incandescent lamp increases and the area under the Intensity- Wavelength curve increases. This verifies Stefan's law